Darrell Long

190/200

1. a) Put the following formula into principle <u>disjunctive</u> normal form:
$(Q \downarrow R) \vee ((\neg P \vee (P \wedge \neg Q)) \wedge R)$

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$(P \wedge \neg Q \wedge R) \vee (P \wedge \neg Q \wedge \neg R) \vee (\neg P \wedge Q \wedge R) \vee (\neg P \wedge \neg Q \wedge R) \vee (\neg P \wedge \neg Q \wedge \neg R)$

<see below>

b) Use a) to write the formula in Σ notation: Σ_{01345}

c) Use b) to write the formula in Π notation: Π_{267}

d) Use c) to write the formula in principle <u>conjunctive</u> normal form:

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$(\neg P \vee \neg Q \vee \neg R) \wedge (\neg P \vee \neg Q \vee R) \wedge (P \vee \neg Q \vee R)$

2. Draw a circuit which will cheaply and efficiently perform the same function as the circuit illustrated below: (show your work and use Karnaugh maps)

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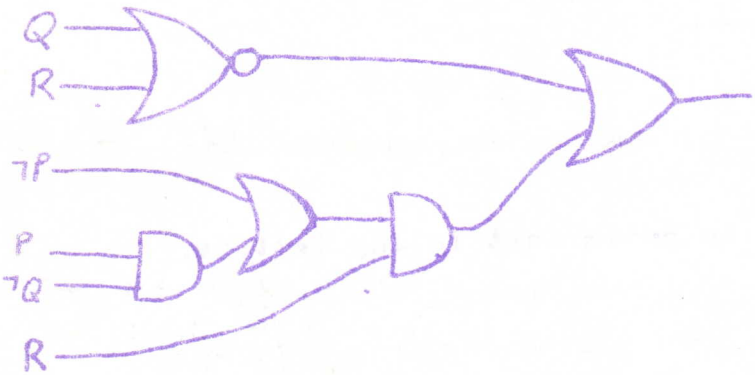
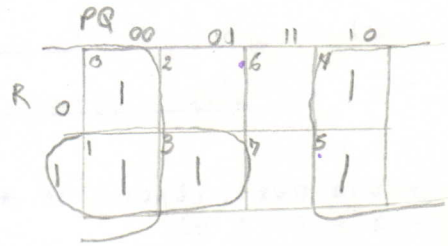
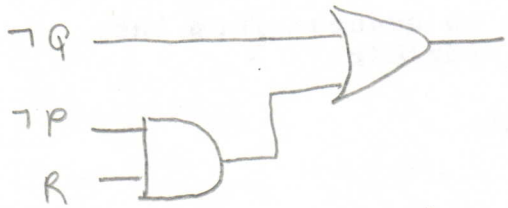




$q' + p'r = \neg Q \vee (\neg P \wedge R)$



1) $(Q \downarrow R) \vee ((\neg P \vee (P \wedge \neg Q)) \wedge R)$

$\Leftrightarrow \neg(Q \vee R) \vee ((\neg P \vee (P \wedge \neg Q)) \wedge R)$ $\quad$ def $\downarrow$

$\Leftrightarrow \neg(Q \vee R) \vee ((\neg P \vee P) \wedge (\neg P \vee \neg Q) \wedge R)$ $\quad$ E_7

$\Leftrightarrow \neg(Q \vee R) \vee (T \wedge (\neg P \vee \neg Q) \wedge R)$

$\Leftrightarrow \neg(Q \vee R) \vee (R \wedge (\neg P \vee \neg Q))$ $\quad$ E_2

$\Leftrightarrow \neg(Q \vee R) \vee (R \wedge \neg P) \vee (R \wedge \neg Q)$ $\quad$ E_6

$\Leftrightarrow (\neg Q \wedge \neg R) \vee (\neg P \wedge R) \vee (\neg Q \wedge R)$ $\quad$ E_3, E_2

$\Leftrightarrow (P \wedge \neg Q \wedge R) \vee (P \wedge \neg Q \wedge \neg R) \vee (\neg P \wedge Q \wedge R) \vee (\neg P \wedge \neg Q \wedge R) \vee (\neg P \wedge \neg Q \wedge \neg R)$

<by addition of missing 'factors'>

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3. Use mathematical induction to show

$n < 2^n$ $\quad P(n): n < 2^n$

$P(0): 0 < 2^0 = 1$ True

$P(m): m < 2^m$

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1) $m+1 < 2^m + 1$; do it to both sides then still true
2) $m+1 < 2^m + 2^m$; $1 < 2^m$
3) $m+1 < 2^1 \cdot 2^m$; def^n of $\cdot$
4) $m+1 < 2^{m+1}$; exponent law

$m+1 < 2^{m+1}$ which is $P(m+1)$, so..

$P(m) \to P(m+1)$, $(\forall x)(P(x) \to P(x+1))$ (UG) and $P(n)$ works for all $n > 0$

4. If f and g are functions whose domain is the set of real numbers, and are given by the formulas at the right, find expressions for the following functions or explain why they do not exist.

$f(x) = 3x^2$

$g(x) = 2x - 4$

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$f \circ g\,(x) =$ $12x^2 - 48x + 48$ $(3(2x-4)^2)$

$g \circ f\,(x) =$ $6x^2 - 4$ $(2(3x^2) - 4)$

$g^{-1}(x) =$ $(x+4)/2$

$f^{-1}(x) =$ dne. since f is not one-to-one, nor onto. $D(f^{-1}) = [0, \infty)$

5. Use the definitions of * and + to prove that in any lattice, $a = a * (a + b)$

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1) $a \le a$ by reflexivity
2) $a \le a + b$ by def^n LUB
3) $a \le a * (a+b)$ by def^n LLB

1) $a * (a+b) \le a$ by def^n of GLB
2) $a \ge a * (a+b)$ def^n $\ge$

so by antisymmetry $a = a * (a+b)$

6. Show that the conclusion $(\exists x)M(x)$ can be deduced from the hypotheses $H_1: (\forall y)(M(y) \vee \neg H(y))$ and $H_2: (\exists z)H(z)$

Identify each implication that you use.

$H_1: (\forall y)(M(y) \vee \neg H(y))$ $\quad H_2: (\exists x)H(x)$ $\quad C: (\exists x)M(x)$

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1) $(\exists x)H(x)$	P
2) $H(a)$	ES (1)
3) $(\forall y)(M(y) \vee \neg H(y))$	P
4) $M(a) \vee \neg H(a)$	US (3)
5) $\neg H(a) \vee M(a)$	E_3 (4)
6) $\neg\neg H(a)$	E_1 (2)
7) $M(a)$	I_{10} (5,6)
8) $(\exists x)M(x)$	EG (7)

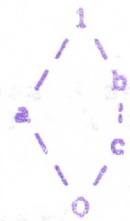
QED

7. In the lattice at the right, what, (if any) are the complement(s) of a? b, c

b? a

Is this a Boolean algebra? no

Why? because the complements are not unique (see a)



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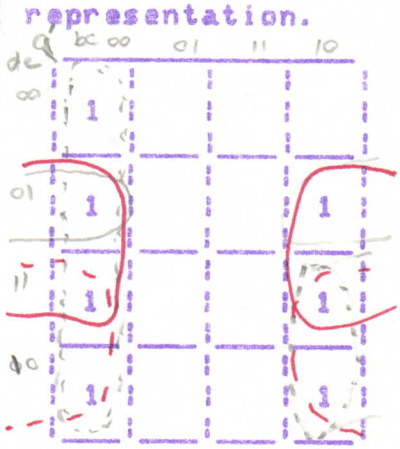
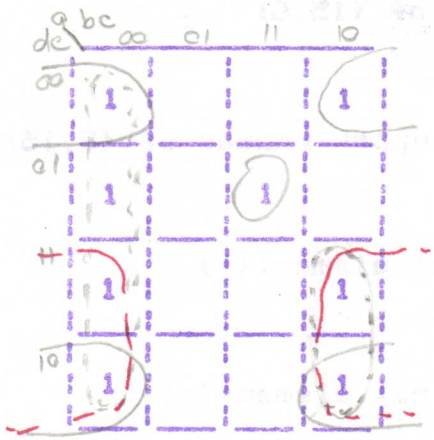
8. Draw in the cubes in the Karnaugh map below and write down the minimal representation.





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$b'c' + ac'e' + a'c'e + bc'd + abcde'$

9. Prove that if a relation R in X is transitive, then $\tilde{R}$ is transitive.

H: $(\forall x)(\forall y)(\forall z)(\langle x,y\rangle \in R \wedge \langle y,z\rangle \in R \rightarrow \langle x,z\rangle \in R)$

C: $(\forall x)(\forall y)(\forall z)(\langle x,y\rangle \in \tilde{R} \wedge \langle y,z\rangle \in \tilde{R} \rightarrow \langle x,z\rangle \in \tilde{R})$

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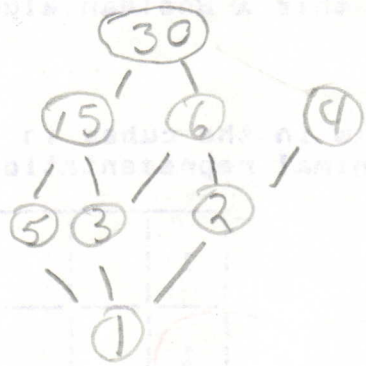
1)	$(\forall x)(\forall y)(\forall z)(\langle x,y\rangle \in R \wedge \langle y,z\rangle \in R \rightarrow \langle x,z\rangle \in R)$	P
2)	$\langle a,b\rangle \in R \wedge \langle b,c\rangle \in R \rightarrow \langle a,c\rangle \in R$	US^3 (1)
3)	$\langle c,b\rangle \in \tilde{R} \wedge \langle b,a\rangle \in \tilde{R}$	assumed
4)	$\langle b,a\rangle \in \tilde{R}$	I_2(3)
5)	$\langle a,b\rangle \in R$	def^n ~
6)	$\langle c,b\rangle \in \tilde{R}$	I_1 (3)
7)	$\langle b,c\rangle \in R$	def^n ~
8)	$\langle a,b\rangle \in R \wedge \langle b,c\rangle \in R$	I_9 (5,7)
9)	$\langle a,c\rangle \in R$	I_{11} (8,2)
10)	$\langle c,a\rangle \in \tilde{R}$	def^n ~
11)	$\langle c,b\rangle \in \tilde{R} \wedge \langle b,a\rangle \in \tilde{R} \rightarrow \langle c,a\rangle \in \tilde{R}$	CP (3,10)
12)	$(\forall x)(\forall y)(\forall z)(\langle x,y\rangle \in \tilde{R} \wedge \langle y,z\rangle \in \tilde{R} \rightarrow \langle x,z\rangle \in \tilde{R})$	UG^3 (11)

QED

10. Draw the Haase diagram for the relation "x is a factor of y" over the set X = {1,2,3,4,5,6,15,30}:



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Find the following (if they exist):

a) the GLB of {15,6} = 3

b) all the upper bounds of {3,15}

= 15, 30

c) the least element(s)

= 1

d) the maximal element(s)

= 30, 4

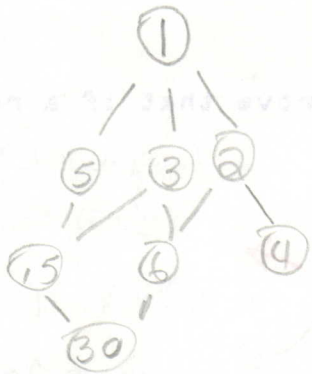
e) the minimal element(s)

= 1

f) the greatest element(s)

d.n.e.

Draw the Haase diagram of the converse relation:



11. Let X = $\mathcal{P}$({b,c}), and let R be the subset relationship in X. Order the 4 elements in X and find M_R:
(Justify your answers.)

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Is R reflexive? yes (1's on diagonal)
Is R symmetric? No ($M_R \neq M_R^T$)
Is R irreflexive? No (It is reflexive)
Is R antisymmetric? Yes ($M_R \wedge M_R^T$ diagonal matrix)
Is R transitive? Yes (M_R^2 "covers" M_R)

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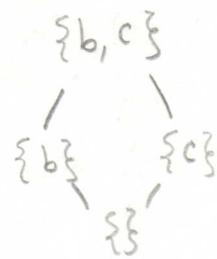
$$M_R = \begin{bmatrix} 1&0&0&1 \\ 0&1&0&1 \\ 0&0&1&1 \\ 0&0&0&1 \end{bmatrix} \quad M_R^T = \begin{bmatrix} 1&0&0&0 \\ 0&1&0&0 \\ 0&0&1&0 \\ 1&1&1&1 \end{bmatrix}$$

$$M_R^2 = \begin{bmatrix} 1&0&0&1 \\ 0&1&0&1 \\ 0&0&1&1 \\ 0&0&0&1 \end{bmatrix} \quad M_R \wedge M_R^T = \begin{bmatrix} 1&0&0&0 \\ 0&1&0&0 \\ 0&0&1&0 \\ 0&0&0&1 \end{bmatrix}$$

	{∅}	{b}	{c}	{b,c}
{∅}	1	1	1	1
{b}	0	1	0	1
{c}	0	0	1	1
{b,c}	0	0	0	1

reflexive